

Homework 1

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Problem 1

$$(a) \ 2u = 2[2, -1, 3]^T = [4, -2, 6]^T$$

$$2u - v = [4 - 1, -2 - 4, 6 + 2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48$$

$$\text{Final answer: } (2u - v) \cdot w = 24$$

$$(b) \ (1/2)w = (1/2)[-4, 2, 6]^T = [-2, 1, 3]^T$$

$$v + (1/2)w = [1 - 2, 4 + 1, -2 + 3]^T = [-1, 5, 1]^T$$

$$||v + (1/2)w|| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27}$$

$$\text{Final answer: } ||v + (1/2)w|| = 3\sqrt{3} \approx 5.20$$

$$(c) \ \text{The angle satisfies } \cos \theta = (u \cdot w) / (||u|| ||w||).$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$||u|| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$$

$$||w|| = \sqrt{(-4)^2 + 2^2 + 6^2} = \sqrt{56} = 2\sqrt{14}$$

$$\text{So } ||u|| ||w|| = \sqrt{14} \cdot 2\sqrt{14} = 28 \text{ and } \cos \theta = 8/28 = 2/7.$$

$$\theta = \arccos(2/7), \text{ and in degrees that is } 73.398...$$

$$\text{Final answer: } \theta \approx 73.40^\circ$$

Problem 2

(a) ~~$AB = [1(2) + 0(1) + (-2)(0), \dots]$~~

AB is undefined. A is 3×3 and B is 2×3 , so A has 3 columns while B has only 2 rows,

and the inner dimensions 3 and 2 do not agree.

BA is defined and is 2×3 , since B is 2×3 and A is 3×3 .

row 1 of BA : $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)]$

row 2 of BA : $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)]$

Final answer: $BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$

(b) Av is $(3 \times 3)(3 \times 1) = 3 \times 1$.

$Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T$

Final answer: $Av = [4, -11, -5]^T$, and VA is undefined because v has 1 column while A has 3 rows ($1 \neq 3$).

(c) Aw is undefined: A has 3 columns and w is 1×3 , so w has only 1 row, $3 \neq 1$.

WA is $(1 \times 3)(3 \times 3) = 1 \times 3$.

$WA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)]$

Final answer: $WA = [4 \ -6 \ -1]$

(d) vw is $(3 \times 1)(1 \times 3)$, so it is 3×3 , and every entry is (entry of v) (entry of w).

Final answer: $vw = \begin{bmatrix} 2 & 0 & -1 \\ -4 & 0 & 2 \\ 6 & 0 & -3 \end{bmatrix}$

wv is $(1 \times 3)(3 \times 1) = 1 \times 1$, the same as the dot product of w and v .

$wv = 2(1) + 0(-2) + (-1)(3) = 2 + 0 - 3$, so $wv = [-1]$.

(e) $A^2 = AA$. Both factors are 3×3 so this is defined.

row 1: $[2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$

$$\text{row 2: } [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$\text{row 3: } [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16, -7]$$

$$\text{Final answer: } A^{-2} = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 16 & -7 \end{bmatrix}$$

(f) B^2 is undefined. B is 2×3 , which is not square, so BB would need the 3 columns

of the first B to match the 2 rows of the second B , and $3 \neq 2$.

Problem 3

$$(a) \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

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Back substitution: $x_3 = 1$, then $x_2 = -4 + 3(1) = -1$, then

$x_1 = -1 - 2(-1) + 1 = 2$. Only one solution since every column has a pivot.

$$\text{Final answer: } x = [2, -1, 1]^T$$

$$(b) \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 2 & 5 & 1 & | & 6 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 2 & 5 & 1 & | & 6 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 2 & 5 & 1 & | & 6 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 2R_1$, then $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$R_3 \rightarrow R_3 - R_2$:

$$[1 \ 2 \ -1 \ | \ 4]$$

$$[0 \ 1 \ 3 \ | \ -2]$$

$$[0 \ 0 \ 0 \ | \ 0]$$

$R_1 \rightarrow R_1 - 2R_2$:

$$[1 \ 0 \ -7 \ | \ 8]$$

$$[0 \ 1 \ 3 \ | \ -2]$$

$$[0 \ 0 \ 0 \ | \ 0]$$

There is no pivot in column 3, so z is a free variable. Put $z = t$.

Then $y = -2 - 3t$ from row 2 and $x = 8 + 7t$ from row 1, so there are infinitely many solutions.

Final answer: $[x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T$ for any $t \in \mathbb{R}$

Problem 4

(a) Let x_1 = mass of Alloy 1 used, in kg; x_2 = mass of Alloy 2 used, in kg;
 x_3 = mass of Alloy 3 used, in kg.

The 100 kg of product contains 36 kg of copper, 44 kg of zinc, and 20 kg of nickel, and no metal is lost, so the mass of each metal in equals the mass of it out.

~~$$0.5x_1 + 0.3x_2 + 0.2x_3 = 36$$~~

copper: $0.5x_1 + 0.2x_2 + 0.4x_3 = 36$

zinc: $0.3x_1 + 0.7x_2 + 0.2x_3 = 44$

nickel: $0.2x_1 + 0.1x_2 + 0.4x_3 = 20$

(b) Scaling each equation by 10 gives whole-number entries:

$$[5 \ 2 \ 4 \ | \ 360]$$

$$[3 \ 7 \ 2 \ | \ 440]$$

$$[2 \ 1 \ 4 \ | \ 200]$$

(c) $R_1 \rightarrow R_1 - 2R_3$:

$$[1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$R_2 \rightarrow R_2 - 3R_1$, then $R_3 \rightarrow R_3 - 2R_1$:

$$[1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$R_2 \rightarrow (1/7)R_2$, then $R_3 \rightarrow R_3 - R_2$:

$$[1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$R_3 \rightarrow (1/10)R_3$:

$$[1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

$x_3 = 20$, so $x_2 = 80 - 2(20) = 40$ and $x_1 = -40 + 4(20) = 40$.

Check: $40 + 40 + 20 = 100$ kg, and copper is $0.5(40) + 0.2(40) + 0.4(20) = 36$ kg.

Final answer: 40 kg of Alloy 1, 40 kg of Alloy 2, and 20 kg of Alloy 3

Problem 5

(a) $R_2 \rightarrow R_2 - 2R_1$, then $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$R_3 \rightarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$R_1 \rightarrow R_1 - R_2$:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Final answer: $\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

(b) $\text{rank}(A) = 2$

(c) Row reducing $[A \mid b]$ with those same operations turns the bottom row into

$$\begin{bmatrix} 0 & 0 & 0 \mid b_3 - b_1 - b_2 \end{bmatrix}.$$

0 solutions: possible. If $b_3 \neq b_1 + b_2$, say $b = [0, 0, 1]^T$, the bottom row reads $0 = 1$, so no x can satisfy the system.

Exactly 1 solution: impossible. Column 2 has no pivot in the RREF, so x_2 is a free variable whenever the system is consistent, and $\text{rank}(A) = 2$ is less than the 3 unknowns. A consistent system therefore always has more than one solution.

Infinitely many: possible. If $b_3 = b_1 + b_2$, say $b = [1, 3, 4]^T$ or $b = 0$, the system

is consistent, and the free variable x_2 produces infinitely many solutions.

Problem 6

(a) Write the vectors as the columns of a matrix and row reduce.

$$R_3 \rightarrow R_3 - 2R_1:$$

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2:$$

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Only two pivots for three vectors, so the homogeneous system has a free variable and the vectors are dependent. Explicitly, $v_2 - 2v_1 = [2, 1, 3]^T - [2, 0, 4]^T = v_3$.

v_1 and v_2 are independent by themselves, so the span is 2-dimensional.

Final answer: linearly dependent, span is a plane through the origin

(b) ~~$R_2 \rightarrow R_2 + R_1$~~

$$R_2 \rightarrow R_2 - R_1:$$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2:$$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$$

Three pivots for three vectors, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ forces $c_1 = c_2 = c_3 = 0$ and the set is independent. Three independent vectors in $\mathbb{R}^3$ span the entire space. Final answer: linearly independent, span is all of $\mathbb{R}^3$